

Online Resource 1: Methods and numerical reproduction

Article: From Set Prediction to Evidence Selection: An Empirical Study of Surrogate Objectives in Multi-hop RAG

Journal: Journal of Intelligent Information Systems

Public neutral reading copy; submission metadata is supplied separately.

Purpose and reading map

This resource documents the existing empirical study; it contains no additional model run, answer scoring or inferential test. It accompanies the complete main article, not an appendix excluded from the article page count. The five main figures and six main tables remain in the main article.

Online Resource 2 contains the saved numerical records and a Python standard-library reconstruction script. Its README maps files to endpoints. Table 1 uses data roles; Table 2 uses retained aggregate reference gaps; Tables 3–5 use saved outcomes and paired transitions; Table 6 uses the refined timing aggregate. Figures 3–5 use the corresponding score bins, transitions and measured times. Case notes support the source-text examples in Figure 1; Figure 2 is a study overview, not an additional result.

The initial static selection used TUNE including QA. Later DEV-SELECT excludes QA, but does not make the 128-question panel an independent test. Both seeds use the same questions. This resource cannot establish independent generalization or repair missing historical records.

Reproduction command and limits

Extract Online Resource 2 and run `python reconstruct.py` in its root. No model package, raw answer file or network connection is required. The derived tables reconstruct existing saved scores; the script does not calculate a new answer score or regenerate a prediction. The reference-gap and refined-cost tables remain retained aggregates, not recovered per-question measurements. The archive includes the full existing de-identified 16-case notes, with their outcome-stratified and assistant-assisted interpretation limits.

Data, configuration, and reconstruction scope

Data sources and roles

- HotpotQA training distractor release v1.1, `hotpot_train_v1.1.json`: official dataset. The study reuses an already evaluated 1,000-question historical cohort.
- MuSiQue-Ans v1.0, `musique_ans_v1.0_train.jsonl`: official repository and data instructions. The study reuses an already evaluated 3,000-question historical cohort.
- The study’s input cohort is not the entirety of either public training set. Obtaining the source releases alone does not reconstruct the historical cohort. Exact historical cohort manifests and legal redistribution/access arrangements are additional prerequisites for a full model rerun; they are not distributed as identifiable records in this anonymous numerical package.
- The resulting 4,000 queries are exposed development data, not a new test set. `data_roles.csv` and `role_overlap.csv` give the actual role counts and query overlaps. FIT and TUNE also share 57 HotpotQA and 4,753 MuSiQue source blocks; question-group separation is not document separation.

Implemented preparation rules

Starting with the historical cohort, join nonempty original sentences in source order into complete titled paragraphs and retain original sentence-to-paragraph spans. HotpotQA supporting-fact coverage remains sentence-based; MuSiQue coverage remains annotated-paragraph-based. Deduplicate paragraphs by dataset, case-folded whitespace-normalized title and text. The two development corpora contain 9,791 and 31,130 blocks.

Union queries sharing an exact normalized question or the same dataset/source question ID. For each group, choose the lexicographically smallest dataset + NUL + normalized-question key. Hash the string `rg-learned-set-v5-split` + NUL + key using SHA-256. A group containing any of the 96 previously exposed interface-development questions has DEV precedence; otherwise the first eight hexadecimal digits modulo five select TUNE when zero, FIT otherwise. Hashes are used here to describe the existing split, not as new integrity tests.

The QA panel is chosen within TUNE by stable group-hash ordering with the existing `v5-panel` salt: 48 bridge/16 comparison HotpotQA and 32 two-hop/32 longer-hop MuSiQue questions. MINE is a stratified 256+256 subset of FIT; DEV_SELECT is a stratified 64+64 subset of TUNE excluding QA groups. Their salts are `rg-v6-mine` and `rg-v6-select`. Quotas use integer proportional allocation with largest-remainder correction and deterministic group order. FIT=3,166 questions/3,165 groups; TUNE=738; DEV=96. The pre-existing DEV membership and historical cohort are required to reproduce this split exactly.

Models, search, and reader

- Encoder: BAAI/bge-large-en-v1.5, revision `d4aa6901d3a41ba39fb536a557fa166f842b0e09`; FP32 CLS pooling followed by L2 normalization, 512-token encoder ceiling. Exact dense Top128 over the relevant development paragraph corpus, no Gold insertion into evaluation candidates.
- Reader: Qwen/Qwen2.5-3B-Instruct, revision `aa8e72537993ba99e69dfaafa59ed015b17504d1`, unadapted FP16. Same tokenizer and constrained answer-field protocol, greedy, maximum 128 new tokens, complete input ceiling 1,024 tokens. Selected whole paragraphs appear in BGE order; no individual paragraph is truncated to force feasibility.
- Learned search: maximum six paragraphs, width-four beam with feasible Dense-K6 included in the final candidates. Only the full-support logit ranks sets. Exact ball/k-means variants preserve this beam procedure, not a global optimum.
- Dense-budget/MMR-budget share the 1,024-token ceiling but may use more than six short blocks. MMR relevance weight is 0.7. K6 variants are used for matched-cardinality support diagnostics.
- Main models H1/H2/H4 and DeepSets share support supervision and query-paragraph features. HOFM factors have rank 32, common representation width 128. Continuation uses two rounds at most, shared 64-set-per-query mined pools, learning rate 0.0003, AdamW weight decay 0.0001, batch eight queries, gradient norm one, and at most eight epochs per round. Checkpoints at epochs 2/4/8 maximize DEV_SELECT completeness, then coverage, then minimize original development loss, then prefer earlier epochs. It changes examples, loss and checkpoint selection together.
- Static H4 replay matches optimizer updates, not all forward evaluations. QA does not choose the continuation checkpoint. Seed repetition uses the same panel; it supplies no additional independent questions.

- The initial static checkpoint used loss on the entire TUNE pool, which includes QA128. Only subsequent DEV_SELECT excludes QA. No claim that QA was untouched by all model selection is made.
- Observed research software: CPython 3.12.0, PyTorch 2.12.1+cu130, Transformers 5.14.1, NumPy 2.5.1, lm-format-enforcer 0.11.3. OMP/MKL/OpenBLAS/NumExpr threads were one, tokenizer parallelism disabled; GPU was RTX 4060 Laptop 8 GB. The numerical reconstruction here requires none of these research packages.

Results to implementation map

Evidence	Supplement input	Reproduction scope
Same-panel Full/F1/EM, both seeds, paired counts	<code>observations.csv</code> , <code>paired_numeric.csv</code>	<code>reconstruct.py</code> , numerical derivation from saved scores (no rescoreing)
Development data roles	<code>data_roles.csv</code> , <code>role_overlap.csv</code>	Existing cohort membership counts, no query IDs distributed
Original selected support	<code>static_selection.csv</code>	Existing aggregate output from paragraph selection evaluation; model rerun not included
Checkpoint-selected support trajectory	<code>checkpoint_selection.csv</code>	Existing aggregate output from shared-pool continuation; not an independent QA result
Actual index timing	<code>index_cost_refined64.csv</code> , <code>index_cost_original738.csv</code>	Distinct refined 32+32 aggregate and original 738-query timing; no new benchmark; nested timing scopes and some CPU/GPU overlap

The experiment implementation separates corpus construction, supervision, scorer, feasible search, continuation training, and reader evaluation. Equations and selection/training rules appear in the manuscripts. Full reruns additionally need the original implementation, cohort manifests, source mappings, authorized labels, checkpoints, and prompt/decoder identity. This package is intentionally sufficient for the reported numerical summaries, not for independently regenerating all model predictions.

Supplementary method and cost notes

These retained method and measured-cost details introduce no new observations, scores, or tests. All numerical inputs retain their prior contents.

Preference availability

MINE contains 256 HotpotQA and 256 MuSiQue FIT questions. The primary-seed shared pool does not supply every preference category for every question; missing categories are masked, while base supervision remains available.

Category / round	HotpotQA unavailable / 256	MuSiQue unavailable / 256
Final pair, round 1	171	115
Frontier pair, round 1	17	42
Final pair, round 2	196	135
Frontier pair, round 2	19	59

These counts are the existing primary-seed reports, not an estimate pooled across seeds. Checkpoints were compared at epochs 2, 4, and 8; all selected aligned models came from round 1. The static third seed is not a continuation replicate.

Historical measured cost ledger

Source: unchanged `cost_static.json` and `cost_aligned.json`. These totals include their original broader development, sensitivity and repeat work, not just the 128-question answer table.

Recorded quantity	Static study	Continuation study
Model-residency GPU process wall seconds	4,350.28	4,513.07
Measured CPU process seconds	10,541.75	7,035.50
New reader calls	2,287	986
Logical answer rows	3,520	1,536
Cache hits	1,241	558
Measured generation seconds	2,532.22	867.94
Peak allocated GPU memory, GiB	6.63	6.11

Each study includes eight reader reruns bypassing inference reuse; new calls plus cache hits exceed logical rows by eight. Cache reuse is neither independent generation nor evidence of deployment acceleration. Dividing these totals by 128 would not estimate a serving cost. Earlier unrecorded CPU time is unknown. Static replay matches optimizer-update count, not the extra evaluations of the aligned preference losses.

The selector comparison in the manuscript instead uses `index_cost_refined64.csv`: 32 fixed questions per dataset, with means and medians. Its per-question timing distribution was not retained. `index_cost_original738.csv` belongs to a separate original 738-question index check. Tokenizer time is nested within total selector time and must not be added again. Some CPU timing overlapped reader execution.

Missing reference records

`reference_gaps.csv` retains the existing two-row aggregate Dense-K6 diagnostic. Per-question Dense-K6 logits and selected sets, and paired MMR-K6 scores, were not saved. No model was loaded to replace them, and no paired score-gap distribution was constructed.

Statements

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